

ABHISHEK REDDY MALREDDY

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EDUCATION

CARNEGIE MELLON UNIVERSITY

GPA: 3.82/4.0

Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering - Materials Science and Engineering

Dec 2025

Selected Coursework: Computer Vision, Systems and Tool Chains for AI Engineers, Trustworthy AI, Deep Learning & LLMs.

NATIONAL INSTITUTE OF TECHNOLOGY CALICUT

Calicut, India

Bachelor of Technology in Mechanical Engineering

May 2022

SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, SQL, C/C++, FastAPI, React, Next.js, Node.js.

AI & ML: LLM integration, RAG, AI agents (LangChain, LangGraph, MCP), PyTorch, Transformers, Whisper, Fine Tuning, Computer Vision, Reinforcement Learning, Graph Neural Networks.

Databases & Infrastructure: PostgreSQL (pgvector, PostGIS), MongoDB, Supabase, Docker, AWS, GCP, RunPod, Git, Linux.

PROFESSIONAL EXPERIENCE

SPAN Enterprises LLC

Fort Mill SC, USA

Junior AI / ML Engineer

Dec 2025 – Present

- Shipped a call analytics platform for a 50 person customer support team handling 9,000+ calls per month, turning recordings into searchable transcripts, summaries, and per representative performance metrics, deployed with Docker on the internal network so call data stays in house.
- Designed the underlying ML pipeline with Whisper transcription, speaker diarization, and a self hosted Qwen 8B LLM to score sentiment, classify topics, and flag risky calls, plus a grounded chat assistant that cites source calls behind each answer.
- Created a central CMS in Python and React powering product content and chatbot knowledge for the live TaxBandits.com and Tax990.com applications, with role based access control and CMS operations exposed as MCP tools.
- Developed a customer facing chatbot and feedback loop on LangChain and LangGraph with agentic tool calling wired to TaxBandits developer APIs, and fine tuned models on RunPod to extract structured data from uploaded tax form images.

VIGILANT INC.

Remote, USA

Artificial Intelligence Engineering Intern

Jun 2025 – Aug 2025

- Designed and deployed a FastAPI microservice powering GENIE ML Deduplicator, the core pipeline ensuring Go Vigilant app users receive unique alerts in real time from 10,000+ global news and intelligence sources.
- Engineered a multistage deduplication pipeline combining static filtering, transformer embeddings, DBSCAN clustering, and semantic similarity with PostgreSQL, pgvector, and PostGIS to detect redundant event coverage and match alerts by location.
- Automated embedding generation with Hugging Face models and containerized the service using a multistage Dockerfile, delivering reproducible deployments powering a production consumer app.

MOBILITY RESEARCH (IIIT-H Affiliated)

Hyderabad, India

Research Assistant **Advisor: Dr. Girish Varma**

May 2023 – Jun 2024

- IDD-AW: Robust Semantic Segmentation for Autonomous Driving in Adverse Conditions:**
 - Developed the large-scale IDD-AW dataset (5000 RGB-NIR images) for semantic understanding of Indian driving scenes under adverse weather (rain, fog, snow, lowlight). Introduced the novel "Safe mIoU" metric to enhance safety evaluation of segmentation models by penalizing critical misclassifications overlooked by traditional mIoU.
 - Built SB3-based reinforcement learning models in SUMO to optimize lane selection and vehicle-to-vehicle communication, reducing emergency vehicle traversal times and surpassing human traffic strategies.
- Published at the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Jan 2024.**

ACADEMIC PROJECTS

Machine Learning Based Characterization of Chemical Ordering in High Entropy Oxides

CMU

Research Assistant **Advisor: Dr. Simon Gelin**

May 2025 – Dec 2025

- Trained equivariant graph neural network potentials (Allegro, NequIP) on DFT data to predict atomic-scale energies and forces, building ML surrogates that replace quantum-mechanical calculations at a fraction of the compute cost.
- Deployed the trained models as the energy engine for Monte Carlo simulations of HEO-J14, enabling sampling of chemical ordering across temperatures at scales infeasible with direct DFT.

TrackAnything: Multi-modal Promptable Tracking in 3D (CMU)

Jan 2025 – Apr 2025

- Built a language-grounded 3D tracking model that maps free-form text prompts (e.g., "walking between parked vehicles") to 3D bounding boxes in driving scenes, enabling text-based scenario mining over the Argoverse 2 dataset.
- Fused text, LiDAR, map, and vehicle pose inputs in a transformer architecture inspired by PromptTrack, aligning language queries with multimodal sensor data.